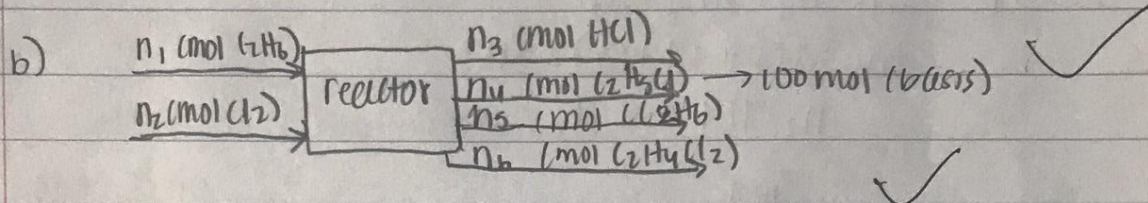


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72

parmyja kotus

a) To maximize selectivity of monochloromethane production, a low ethane conversion with an excess of ethane. It allows the reaction to proceed by creating more of the monochloromethane while minimizing the amount of dichloromethane created in the side rxn.



DOF: 4a: 5 5a: 1 7-5=2 unspecified

4b: 2 5b: 0

7

5c: 0

5d: 4
5

$$\frac{14 \text{ mol C}_2\text{H}_5\text{Cl}}{1 \text{ mol C}_2\text{H}_4\text{Cl}_2} = \frac{100 \text{ mol C}_2\text{H}_5\text{Cl}}{x \text{ mol C}_2\text{H}_4\text{Cl}_2}$$

↑
 n_6

c) atomic balance on reactor: $In = Out$

$$n_6 = 7.143 \text{ mol C}_2\text{H}_6$$

$$\text{H: } 6n_1 = 1n_3 + 5n_4 + 6n_5 + 4n_6$$

$$\text{C: } 2n_1 = 2n_4 + 2n_6 + 2n_5$$

$$\text{Cl: } 2n_2 = 1n_4 + 2n_6 + 1n_3$$

$$n_4 = 100 \text{ mol C}_2\text{H}_5\text{Cl}$$

$$n_5 = (1 - 0.15)n_1 \rightarrow n_5 = (0.85)(714)$$

$$n_6 = 7.143 \text{ mol C}_2\text{H}_6$$

$$n_5 = 607.14 \text{ mol C}_2\text{H}_6$$

$$\text{C: } 2n_1 = 2n_4 + 2n_6 + 2n_5$$

$$2n_1 = 2(100) + 2(7.143) + 2(0.85)n_1$$

$$2n_1 = 214.286 + 2n_1 - 0.3n_1$$

$$n_1 = 714 \text{ mol C}_2\text{H}_6$$

$$H: 6n_1 = n_3 + 5n_4 + 6n_5 + 4n_6$$

$$6(7.14) = n_3 + 5(100) + 6(607.14) + 4(7.143)$$

$$n_3 = 113 \text{ mol HCl}$$

$$Cl: 2n_2 = n_4 + 2n_6 + n_3$$

$$2n_2 = 100 + (7.143)2 + 113$$

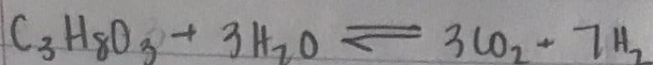
$$n_2 = 114 \text{ mol Cl}_2$$

$$\text{Frac yield: } \frac{\text{mol desired prod}}{\text{mol LR}} = \frac{100 \text{ mol C}_2\text{H}_5\text{Cl}}{114 \text{ mol Cl}_2} = .877 = 87.7\%$$

$$\text{feed ratio: } \frac{114 \text{ mol Cl}_2}{714 \text{ mol tot.}} = .1596 = \boxed{15.96\%}$$

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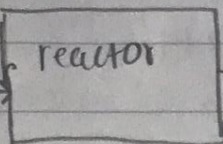
4.4



$$\dot{n}_G = 150 \frac{\text{g}}{\text{hr}} = 1.66 \frac{\text{mol}}{\text{hr}}$$

$$\dot{n}_W = 5(150) \frac{\text{g}}{\text{hr}} = 8.3 \frac{\text{mol}}{\text{hr}}$$

$$\dot{n}_{\text{total, in}} = 9.96$$



$$\dot{n}_G =$$

$$\dot{n}_W =$$

$$\dot{n}_H =$$

$$\dot{n}_C =$$

$$\dot{n}_{\text{total, out}} =$$

gas phase : 1.2 atm & 200°C

water = 5 mol

$$\text{glycerin} = 1 \text{ mol} \rightarrow \frac{150 \text{ g/hr}}{92 \text{ g/mol}} = 1.66 \frac{\text{mol}}{\text{hr}}$$

glycerin molar flow rate = 150 g/hr

$$\text{C}_3\text{H}_8\text{O}_3: \dot{n}_{G, \text{out}} = 1.66 \frac{\text{mol}}{\text{hr}} - \dot{\xi}_1 \rightarrow \dot{n}_{G, \text{out}} = 1.66 - 1.4$$

$$\dot{n}_{G, \text{out}} = .26 \text{ mol/hr}$$

$$\text{H}_2\text{O}: \dot{n}_{W, \text{out}} = \dot{n}_{W, \text{in}} - 3\dot{\xi}_1 \rightarrow 5(1.66 \frac{\text{mol}}{\text{hr}}) - 3(1.4)$$

$$\dot{n}_{W, \text{out}} = 4.1 \frac{\text{mol}}{\text{hr}}$$

$$\text{CO}_2: \dot{n}_{C, \text{out}} = \dot{n}_{C, \text{in}} + 3\dot{\xi}_1 \rightarrow 3(1.4) \rightarrow \dot{n}_{C, \text{out}} = 4.2 \frac{\text{mol}}{\text{hr}}$$

$$\text{H}_2: \dot{n}_{H, \text{out}} = \dot{n}_{H, \text{in}} + 7\dot{\xi}_1 \rightarrow 7(1.4) \rightarrow \dot{n}_{H, \text{out}} = 9.8 \frac{\text{mol}}{\text{hr}}$$

cons

total

feed

rate

$$.54 = \frac{7\dot{\xi}_1}{(1.66 - \dot{\xi}_1) + (8.3 - 3\dot{\xi}_1) + (3\dot{\xi}_1 + 7\dot{\xi}_1)} \rightarrow \dot{\xi}_1 = 1.4 \rightarrow \text{plug back in}$$

$$\dot{n}_{\text{total out}} = .26 + 4.1 + 4.2 + 9.8 = 18.36 \text{ mol/hr}$$

$$\text{Fractional conversion} = \frac{1.66 - .26}{1.66} = .843 \rightarrow 84.3\% \quad \checkmark$$

$$K_a = \frac{[X_{CO_2}]^3 [X_{H_2}]^7}{[X_{C_3H_8O_5}] [X_{H_2O}]^3} \cdot p^6$$

$$X_G = \frac{n_{G,out}}{n_{total}} = \frac{0.26}{18.36} = 0.014$$

$$X_N = \frac{n_{N,out}}{n_{total}} = \frac{4.1}{18.36} = 0.2233$$

$$X_C = \frac{n_{C,out}}{n_{total}} = \frac{4.2}{18.36} = 0.2287$$

$$X_H = \frac{n_{H,out}}{n_{total}} = \frac{9.8}{18.36} = 0.5334$$

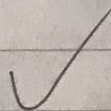
using numerical methods $\rightarrow$ goal seek $\rightarrow$ find ξ_1 , the extent of reaction

$$\xi_1 = 1.57$$

$$K_a = \frac{[0.2287]^3 [0.5334]^7}{[0.014] [0.2233]^3} \cdot (1.2)^6$$

$K_a = 3.8 \text{ atm}^6 \rightarrow 55 \text{ atm}^6 \gg 3.8 \text{ atm}^6 \therefore$ the reaction is not at equilibrium

$\rightarrow$ in order to achieve equilibrium, one would change the temperature and/or pressure. The temperature and pressure effect concentrations, thus affecting the equilibrium constant. could be adjusted accordingly to achieve equilibrium.



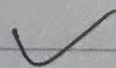
2.

new mol fraction H_2 @ equilibrium

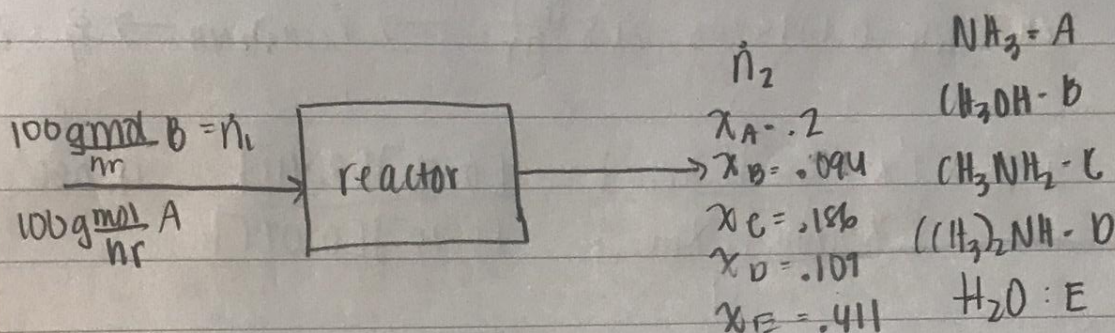
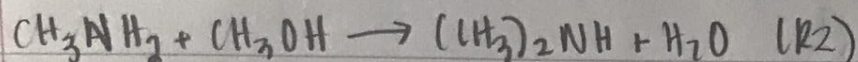
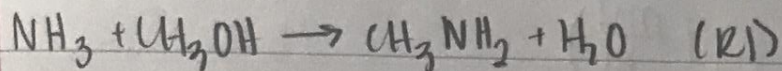
$$X_{H_2} = \frac{n_{H_2, out}}{n_2} = \frac{7 \dot{F}_1}{9.81 + 6 \dot{F}_1} = \frac{7(1.57)}{9.81 + 6(1.57)} = 0.57 \frac{\text{mol } H_2}{\text{total mol}}$$

new fractional conversion @ equilibrium

$$\frac{(1.66) - (1.66 - 1.57)}{(1.66)} = 0.945 \rightarrow 94.5\% \text{ conversion}$$



4.



$$\begin{aligned}
 n_{\text{out}} &= n_{A,\text{in}} + n_{B,\text{in}} + v_1 \dot{\xi}_1 + v_2 \dot{\xi}_2 \\
 &= 100 + 100 + \dot{\xi}_1 (-1 -1 +1 +1) + \dot{\xi}_2 (-1 -1 +1 +1) \\
 &= 200
 \end{aligned}$$

atomic balance

$$\text{C: } 100 = (0.094 + 0.186 + (2 \times 0.186) \times 200) \rightarrow 100 \approx 99.8$$

$$\begin{aligned}
 \text{H: } 4(100) + 3(100) &= (4 \times 0.094) + (3 \times 0.20) + (5 \times 0.186) + (7 \times 0.109) \\
 &\rightarrow 100 \approx 99.6
 \end{aligned}$$

$$\text{O: } 100 = (0.094 + 0.411) 200 \rightarrow 100 \approx 101 \quad \checkmark$$

$$\text{N: } 100 = (0.20 + 0.186 + 0.109) 200 \rightarrow 100 \rightarrow 99.8 \quad \checkmark$$

$$\text{fractional conversion} = \frac{\text{mol cons} - \text{mol prod}}{\text{mol con}} = \frac{100 - (0.094 \times 200)}{100} = 0.81 \quad \checkmark$$

$$\text{yield} = \frac{\text{moles of desired prod formed}}{\text{moles of LR fed}} = \frac{0.186 \times 200}{100 - (0.094 \times 200)} = 0.37 \quad \checkmark$$

$$\text{selectivity} = \frac{\text{moles desired prod formed}}{\text{moles of LR cons}} = \frac{0.186 \times 200}{100} = 0.46 \quad \checkmark$$

$$\text{rxn: } n_{\text{outA}} = n_{\text{in}} - \dot{\xi}_1$$

$$.2(200) = 100 - \dot{\xi}_1$$

$$\dot{\xi}_1 = 60 \quad \checkmark$$

P 4.49 -16

reaction rate : $\dot{r}_A (\frac{\text{mol}}{\text{hr}})$

$$(\text{CH}_3)_2\text{NH} = 0.109 \times 200 = 21.8 \frac{\text{mol}}{\text{hr}} \rightarrow \dot{r}_{(\text{CH}_3)_2\text{NH}} \rightarrow \dot{\xi} = 21.8$$

$$K_{A1} = \frac{[\text{X}_{\text{CH}_3\text{NH}_2}][\text{X}_{\text{H}_2\text{O}}]}{[\text{X}_{\text{CH}_3\text{OH}}][\text{X}_{\text{NH}_2}]} = \frac{[0.186][0.411]}{[0.20][0.094]} = 4.066 \approx 4 \checkmark$$

= 4

$$K_{B1} = \frac{[\text{X}_{(\text{CH}_3)_2\text{NH}}][\text{X}_{\text{H}_2\text{O}}]}{[\text{X}_{\text{CH}_3\text{NH}_2}][\text{X}_{\text{CH}_3\text{OH}}]} = \frac{[0.109][0.411]}{[0.186][0.094]} = 2.56 \approx 2.5 \checkmark$$

= 2.5

$$K_{A1} = \frac{[\text{X}_{\text{CH}_3\text{NH}_2}][\text{X}_{\text{H}_2\text{O}}]}{[\text{X}_{\text{CH}_3\text{OH}}][\text{X}_{\text{NH}_2}]} \rightarrow 4 = \frac{(\dot{\xi}_1 - \dot{\xi}_2)(\dot{\xi}_1 + \dot{\xi}_2)}{(\dot{n}_{\text{CH}_3\text{OH}, \text{in}} - \dot{\xi}_1 - \dot{\xi}_2)(\dot{n}_{\text{NH}_2} - \dot{\xi}_1 - \dot{\xi}_2)}$$

=

$$K_{B1} = \frac{[\text{X}_{(\text{CH}_3)_2\text{NH}}][\text{X}_{\text{H}_2\text{O}}]}{[\text{X}_{\text{CH}_3\text{NH}_2}][\text{X}_{\text{CH}_3\text{OH}}]} = 2.5 = \frac{(\dot{\xi}_2)(\dot{\xi}_1 + \dot{\xi}_2)}{(\dot{\xi}_1 - \dot{\xi}_2)(\dot{n}_{\text{CH}_3\text{OH}, \text{in}} - \dot{\xi}_1 - \dot{\xi}_2)}$$

$\dot{n}_{\text{NH}_4, \text{in}}$	$\dot{n}_{\text{CH}_3\text{OH}}$	$\frac{\dot{n}_{\text{NH}_4}}{\dot{n}_{\text{CH}_3\text{OH}}}$	$\dot{\xi}_1$	$\dot{\xi}_2$	% CH_3NH_2	$\frac{\text{CH}_3\text{NH}_2}{(\text{CH}_3)_2\text{NH}}$	$\frac{[\text{X}_{\text{NH}_4}]}{[\text{X}_{\text{CH}_3\text{NH}_2}]}$
100	50	2	36.8	8.1	19	3.5	0.29
100	100	1	59.4	21.8	18.8	1.7	0.376
100	200	.5	85.7	47.4	11.7	0.75	0.36
100	400	.25	95.7	74.0	4.4	0.29	0.22

In order to achieve better selectivity towards CH_3NH_2 , there must be excess NH_4

CH_3OH

0.37

0.37

0.15

0.09

